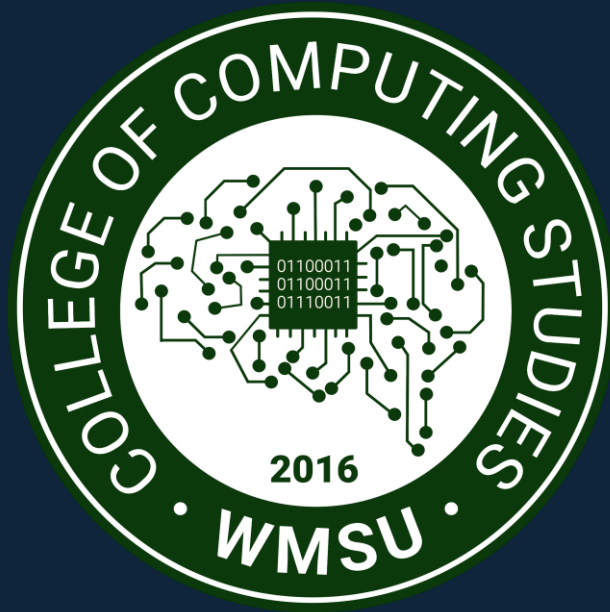




**COLLEGE OF
COMPUTING STUDIES**

IT 140

Capstone Project and Research 1

JASON CATADMAN

Writing Chapters 1, 2, and 3 of the Research Proposal

Components of a Capstone Project Proposal

Capstone Project and Research



Chapter I – Introduction

Chapter II – Review of Related Literature

Chapter III – Conceptual Framework / Technical
Background

Chapter IV – Design and Methodology

CHAPTER 1

Capstone Project and Research

Chapter I – Introduction (Project Context)

1. Project Context
2. Purpose and Description
3. Project Objectives
 - I. General Objectives
 - II. Specific Objectives
4. Scope and Limitations
5. Significance of Study
6. Definition of Terms

Capstone Project and Research

1. Project Context

(must be at least 2 pages of presentation and discussions of...)

1. The proponent should introduce the presentation of the problem, that is, what is the problem is all about. The proponent should describe the existing and prevailing problem situation based on his or her experience. This scope may be global, national, or regional and local.

Capstone Project and Research



2. The proponent should give strong justification for selecting such research problem in his/her capacity as a researcher. Being part of the organization or systems and the desire and concern to improve the systems.
3. The researcher state a sentence or two that would show the link and relationship of the rationale of the study to the proposed research problem.

Capstone Project and Research

Suggested Content:

- Paragraph 1 - Introduce your topic
- Paragraph 2 - Establish the research focus
- Paragraph 3 - State the specific problem
- Paragraph 4 - Describe your study and possible gains/outputs
(The solution)

Capstone Project and Research

2. Purpose and Description of the Project

(must be at least 3-4 paragraphs discussions, each paragraph at least 4 sentences)

What is the function of your project?

What is good in your project?

What makes your project unique, innovative, and relevant?

Capstone Project and Research

3. Project Objectives

Guidelines in Formulating the Objectives of the Project:

1. Start with the **General Objective** which is very parallel to the project title. (*must be at least 2-3 paragraphs*)
2. Explode the general objective into **Specific Objectives** that will help realize the proposed study.
3. Objectives should be **SMART** (*please see introduction to capstone project and research, SMART topic*)

Capstone Project and Research

4. Scope and Limitations of the Project

(must be at least 2-3 paragraphs discussions, each paragraph at least 4 sentences)

- Think the project scope as a **box**. High-level scope defines the sides of the box and separates what is relevant to your project from what is irrelevant.
- The **scope** refers to the work that needs to be accomplished to deliver a product, service, or result with the specified features and functions.
- The **scope** explains the nature, coverage, and time frame of the study

Capstone Project and Research

- The limitation, on the other hand, explains all that are **NOT** included in your project.
- In other words, the scope of the project gives an overview all the **deliverables (i.e. the things that your project gives/delivers)**, and the **tools and technologies** used that will be used in the project development while the limitations of the project are the **boundaries** of the project (i.e. areas/things that are out of scope).

Capstone Project and Research

5. Significance of the Study

Significance of the Study refers to the section of a capstone project or research paper that explains the importance and potential impact of the research. This section addresses why the study is worth conducting and how it contributes to the existing body of knowledge. It often includes:

Theoretical Significance: How the study adds to, supports, or challenges existing theories or frameworks within the field.

Capstone Project and Research



Practical Significance: The real-world applications and benefits of the research findings. This might include implications for industry practices, policy changes, or improvements in professional practices.

Social Significance: The broader societal impact of the research. This could involve contributions to public welfare, community development, or addressing social issues.

Academic Significance: How the study advances academic knowledge, fills research gaps, or opens avenues for further research.

Capstone Project and Research

6. Definition of Terms

Definition of Terms is a section in a capstone project or research paper that provides clear and precise definitions of key terms and concepts used throughout the study. This section ensures that readers understand the specific meanings of terms as they are applied within the context of the research. It typically includes:

Technical Terms: Definitions of specialized terms or jargon that are specific to the field of study.

Capstone Project and Research



Operational Definitions: Specific explanations of how abstract concepts or variables are measured or identified within the research. These definitions are crucial for ensuring clarity and consistency in how terms are used and interpreted.

Context-Specific Definitions: Clarifications of terms that might have multiple meanings or interpretations, to specify how they are being used in the context of the study.

CHAPTER 2

Capstone Project and Research

Chapter II – Review of Related Literature

A survey or review of related literature and studies is very important because such reviews literature and studies serve as a foundation of the proposed study. This is because related literature and studies guide the researcher in pursuing his research venture.

Capstone Project and Research



The following are the different ways on how the review of related literature and studies help as a guide to the researcher:

- They help or guide the researcher in searching for or selecting a better research problem or topic.
- They help the investigator understand his topic for research better.
- They ensure that there will be no duplication of other studies.

Capstone Project and Research



- They help and guide the researcher in locating more sources of related information.
- They help the researcher in making his research design.
- They help and guide the researcher in making comparison between his findings with the findings of other researchers on similar studies with the end in view of formulating generalizations or principles which are the contributions of the study to the fund of knowledge.

Capstone Project and Research

Characteristics of Related Literature and Studies

- The surveyed materials must be as recent as possible.
- Materials reviewed must be objective and unbiased.
- Materials surveyed must be relevant to the study.
- Surveyed materials must have been based upon genuinely original and true facts or data to make them valid and reliable.
- Reviewed materials must not be too few or too many.

Capstone Project and Research

Sources of Related Literature and Studies

- Books, encyclopedias, almanacs, and other similar references
- Articles published in journals, magazines, periodicals, newspapers, and other publications.
- Manuscripts, monographs, memoirs, speeches, letters, and diaries
- Unpublished theses and dissertations

Capstone Project and Research



- The Constitution, and laws and statutes of the land
- Bulletins, circulars, and orders emanating from government offices and departments, especially from the Office of the President of the Philippines and the Department of Education
- Records of schools, public and private, especially reports of their activities
- Official reports of all kinds, educational, social, economic, scientific, technological, political, etc. from the government and other entities
- Articles from the Internet

Capstone Project and Research

Where to locate related literature and studies?

- Libraries, either government, school or private libraries
- Government and private offices
- The National Library
- The Library of Education
- The Internet

Capstone Project and Research

Review of Related Literature

This portion of the proposal manuscript contains presentations and discussions of the following two (2) components:

1. Related Theories
2. Related Projects

Capstone Project and Research

Guidelines in Writing the Review of Related Systems

1. Related Theories

1. Outline first, starting off with an anchor theory
2. Supporting theories help elaborate the anchor theory
3. End noting and footnoting is important which follows correct bibliography entry
4. Fluidity and continuity should be observed

Capstone Project and Research

2. Related Projects

1. Overview of the current system/project
2. Inventory of every related and existing projects/systems
3. Fluidity and continuity should be observed
4. Comparative matrix may be more appropriate
5. Screen shots help make the presentation believable
6. May consider 3 to 6 related studies/projects

Capstone Project and Research

References.

Include all references cited in your proposal, including websites (Chicago style is recommended). Below is an example of how to format different types of sources in Chicago style:

Books

Author's Last Name, First Name. Title of Book. Place of Publication: Publisher, Year of Publication.

Example:

Smith, John. Understanding Research Methods. New York: Oxford University Press, 2020.

Capstone Project and Research

Journal Articles

Author's Last Name, First Name. "Title of Article." Title of Journal volume number, no. issue number (Year of Publication): page numbers. DOI or URL if available.

Example:

Doe, Jane. "The Impact of Technology on Education." Journal of Educational Research 15, no. 2 (2021): 123-145. <https://doi.org/10.1080/12345678.2021.1234567>.

Websites

Author's Last Name, First Name. "Title of Webpage." Name of Website. Last modified Date. URL.

Example:

Brown, Lisa. "The Future of Artificial Intelligence." Tech Innovations. Last modified March 5, 2023. <https://www.techinnovations.com/future-of-ai>.

CHAPTER 3

Capstone Project and Research

Chapter III – Conceptual Framework and Technical Background

2.1 Conceptual Framework

- **Introduction to the Framework**

- Brief discussion of the system concept.
- Reference to related theories or models (e.g., IPO, waterfall, agile, systems theory, etc.)

- **Framework Diagram**

- Input–Process–Output (IPO) Model or equivalent.

Capstone Project and Research

Explanation of Components

- What are the Inputs?
- What happens during the Process?
- What are the expected Outputs?

Research Paradigm

- Visual + textual explanation showing flow of data, user interaction, and outcomes.

Capstone Project and Research

2.2 Technical Background

Current Technologies Used (if upgrading or replacing an existing system)

- Hardware, software, and network specifications.

Proposed Technologies

- Language, frameworks, tools, and platforms to be used (e.g., Python, OpenCV, MySQL, Flask/Django).

Hardware Requirements

- Cameras, servers, computing devices.

Software Stack

- Backend, frontend, database, third-party libraries, etc.

Capstone Project and Research

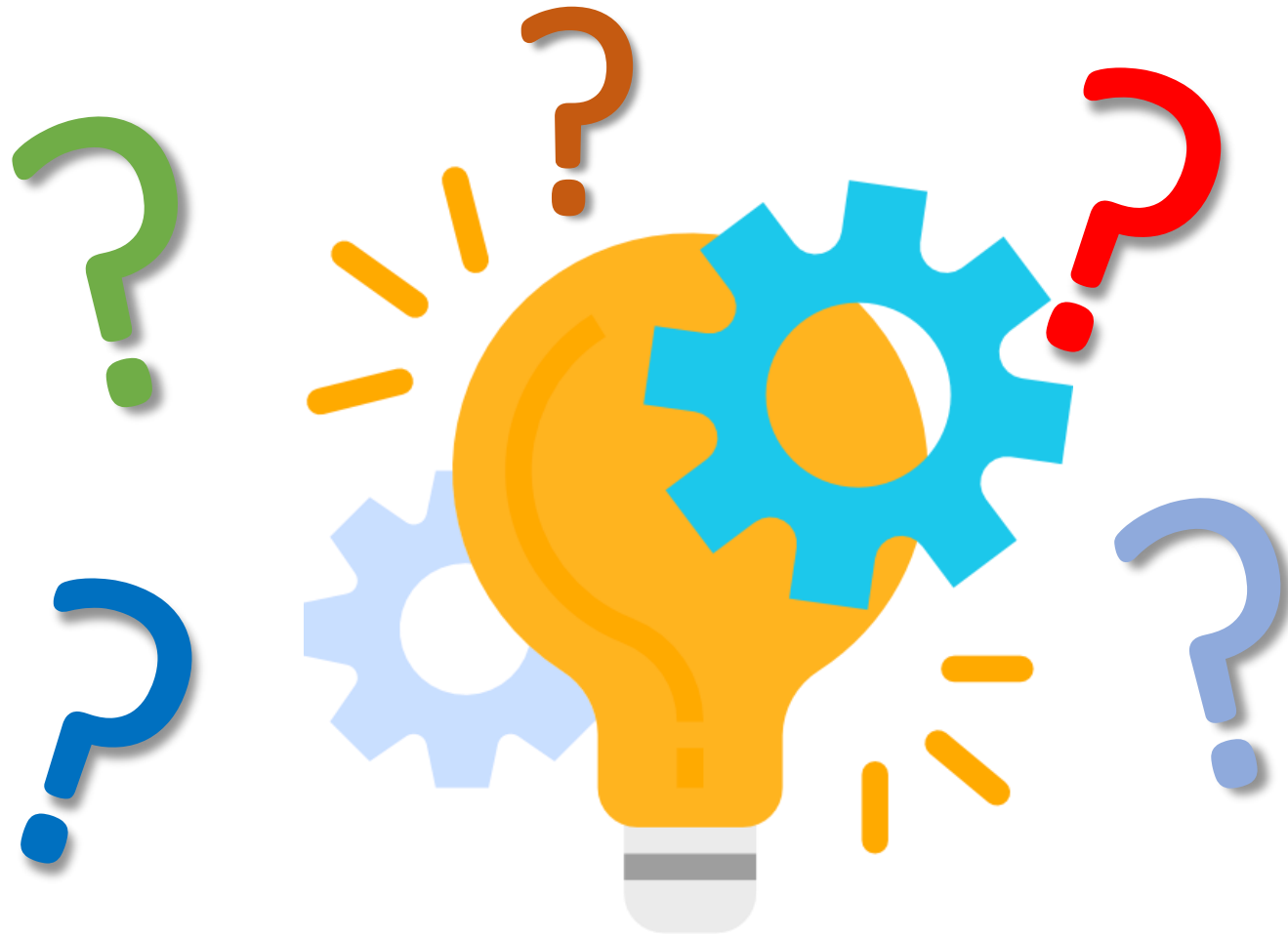


Network Architecture

- If applicable (LAN, cloud, real-time streaming).
- Peopleware
- Who will interact with the system (admins, traffic enforcers, analysts).

Technology Trends and Justification

- Why ML, computer vision, and smart systems are relevant today.



QUESTIONS?



Keep moving Forward.. .